Financial Mathematics

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Tamirat T. Dufera(PhD)

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Adama Science and Technology University
Applied Mathematics Department



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Basic Finance

Interest

- Interest
  - ✓ is a fee paid by one party for the use of assets of another.
  - ✓ It is time dependent



Simple Interest

- Simple Interest:
  - ✓ Interest is paid only on the initial deposit
  - ✓ Let A_0 be initial deposit. After t years the account has value

$$A_t = A_0(1 + tr) \tag{1}$$

here $r \times 100\%$ is the annual rate.



Discrete-Time Compound Interest

- Interest is paid not just on the principal A_0 but on the accrued amounts in the account.
  - ✓ pays annual rate $r \times 100\%$ but with interest compounded m times per year,
    - monthly $(m = 12)$,
    - quarterly $(m = 4)$, or
    - daily $(m = 365)$,
  - ✓ The interest rate per period is then $i := r/m$
  - ✓ The value of the account after t years is

$$A_t = A_0(1 + r/m)^{mt} \tag{2}$$

  - ✓ In this context, A_0 is called the present value or discount value of the account and A_t is the future value at time t.



Continuous-Time Compound Interest

- Write (2) as

$$A_t = A_0 \left[(1 + 1/x)^x\right]^{rt}, \quad \text{where } x = m/r.$$

As $m \to \infty$, $(1 + 1/x)^x \to e$, hence

$$A_t = A_0 e^{rt}. \tag{3}$$



Problem 1

Compare the interest methods for initial deposit $A_0 = 1000$, $r = 3.5\%$, *time* 5.4 *years. Make the comparison of the future value for simple,* $m = 12$, 365 *and continuous.*



Effective Rate

- Effective rate is defined as the simple interest rate that produces the same yield in one year as compounded interest
- Thus if interest is compounded m times a year, then the effective rate must satisfy the equation

$$r_e = (1 + \frac{r}{m})^m - 1.$$



Inflation

- Inflation is defined as an increase over time of the general level of prices of goods and services, resulting in a decrease of purchasing power.
- Inflation is governed by a compound interest formula.
- If r_f is the inflation at annual rate and the price of an item is A_0 at current time, then the price
  - ✓ after one year is $A_0(1 + r_f)$,
  - ✓ after two years $A_0(1 + r_f)^2$, and so on.



Example 2

If $r_f = 25\%$ per year. A 100 Birr price now will be $100(1 + 0.25)^2 = 156.25$ Birr after two years.



Inflation-adjusted rate of return

- The combined effect of inflation and interest.
- Let r be a yearly compounded rate of return on an investment. The actual annual rate of return r_a with inflation taken into account is

$$r_a = \frac{1+r}{1+r_f} - 1 = \frac{r - r_f}{1+r_f}$$

- For small r_f, $\quad r_a \approx r - r_f$



- Let r be the annual interest rate for an investment paying Q dollars in one months.
- Considering the inflation rate r_f, the current value of this payment is

$$A_0 = \frac{1}{1 + i - i_f} Q$$

where $i = \frac{r}{12}, \quad i_f = \frac{r_f}{12}$.



- If a payment of Q two months form now would have time-1 value

$$(1+i-i_f)^{-1}Q$$

and hence time-0 value of

$$A_0 = (1+i-i_f)^{-2}Q$$

- In general if you are to receive a payment of Q in n months, then the present value of this payment, taking monthly inflation into account, is

$$A_0 = (1+i-i_f)^{-n}Q$$



Annuities

- An annuity is a sequence of periodic payments of a fixed amount P.
- Examples: the payment sequences of:
  - ✓ Social Security funds
  - ✓ pensions
  - ✓ car loans,
  - ✓ credit card dept, and
  - ✓ mortgages



- Assume interest at an annual rate r compounded m times per year and that a deposit (withdrawal) is made at the end of each compounding interval.
- The value A_n (the value immediately after the nth deposit (withdrawal)) of the account at time n will be computed



Deposits (Sinking fund)

Theorem 3
At the end of n periods, the future value of the simple ordinary annuity(payments occur at the end of each period) with payment P and $m-$ periodic compounding at r per annum is:

$$A_n = P\frac{(1+i)^n - 1}{i}, \quad i := \frac{r}{m}. \tag{4}$$

More generally, if a deposit of A_0 is made at time 0, then the time-n value of the account is

$$A_n = A_0(1+i)^n + P\frac{(1+i)^n - 1}{i},$$

Proof.

□



Withdrawals

- Let A_0 be the initial value of the account and let A_n denote the value of the account immediately after the nth withdrawal.
- The value just before withdrawal is A_{n-1} plus the interest over that period.
- Show that

$$A_n = (1+i)^n A_0 + P\frac{1-(1+i)^n}{i} \tag{5}$$



- The initial deposit required to support exactly N withdrawals of size P form say, a retirement account or trust fund is given by

$$A_0 = P\frac{1-(1+i)^{-N}}{i} \tag{6}$$

- or solving for P,

$$P = A_0\frac{i}{1-(1+i)^{-N}} \tag{7}$$

  which may be used, for example to calculate the mortgage payment for a mortgage of size A_0.



- Substitute (7) into (5) we get

$$A_n = A_0 \frac{1-(1+i)^{n-N}}{1-(1+i)^{-N}} \tag{8}$$

the time-n value of an annuity supporting exactly N withdrawals.



Application: Retirement

Retirement account:

- make monthly deposits of size P
- an annual rate r, compounded monthly
- After t years transfer the account to one that pays annual rate of r' compounded monthly.
- monthly withdrawals of size Q for s years, drawing down the account to zero



- Final value of the first account is the initial of the second

$$P\frac{(1+i)^{12t}-1}{i} = Q\frac{1-(1+i')^{-12s}}{i'}, \quad i = \frac{r}{12}, i' := \frac{r'}{12} \tag{9}$$

- Solving for P we obtain the formula

$$P = \frac{i}{i'}\frac{1-(1+i')^{-12s}}{(1+i)^{12t}-1}Q \tag{10}$$

- Taking into account that yearly inflation $r_f \times 100\%$, one can get

$$\frac{P}{Q} = \left(\frac{i(1+i_f)^{12t}}{(1+i)^{12t}-1}\right)\left(\frac{1-(1+i_f)^{12s}(1+i')^{-12s}}{i'-i-f}\right).$$



Application : Amortization

- Suppose a home buyer takes out a mortgage in the amount A_0 for t years at annual rate r, compounded monthly.
- From the point of view of the mortgage company, the monthly mortgage payments P constitute a withdrawal annuity with “initial deposit" A_0 and is given by

$$P = A_0 \frac{i}{1-(1+i)^{-N}}, \quad i = \frac{r}{12}, N = 12t$$

- The amount A_n still owed by the homeowner at the end of month n is given by

$$A_n = A_0 \frac{1-(1+i)^{n-N}}{1-(1+i)^{-N}}$$



- Let I_n (Interest) and P_n(Principal) be portions of nth payment

$$I_n = iA_{n-1} \quad \text{and} \quad P_n = P - I_n$$

which implies,

$$I_n = iA_0 \frac{1-(1+i)^{n-1-N}}{1-(1+i)^{-N}} \quad \text{and} \quad P_n = iA_0 \frac{(1+i)^{n-1-N}}{1-(1+i)^{-N}}$$

- The sequences $(A_n), (P_n)$, and (I_n) for what is called an amortization schedule



- Note that

$$\frac{I_n}{P_n} = \frac{1-(1+i)^{n-1-N}}{(1+i)^{n-1-N}} = \frac{(1+i)^{N+1}}{(1+i)^n} - 1$$

- The ratio decreases as n increases
- The maximum and minimum ration are given by

$$\frac{I_1}{P_1} = (1+i)^N - 1 \quad \text{and} \quad \frac{I_N}{P_N} = i$$

  respectively.
- Demonstrate an example using software.



Continuous Income Stream

- Let an account pays a nominal interest rate r receives money at a rate of $f(t)$ dollars per year over T years.
- form t to $t + \Delta t$, the account receives approximately $f(t)\Delta t$ dollars
- The present value is $e^{-rt} f(t)\Delta t$
- Summing these amount and taking $\Delta t \to 0$, we get $\int_0^T e^{-rt} f(t)dt$
- Hence, the present value of an income stream over T years:

$$P = \int_0^T e^{-rt} f(t)dt.$$



Bonds

- A bond is a financial contract issued by: governments, corporations or other institutions.
- It requires that the holder be reimbursed a specific amount at a prescribed time in future.



- The purchaser of a bond pays B_0 and receives a prescribed amount F

$$B_0 = Fe^{-rT}$$

  - ✓ F is called face value or par value,
  - ✓ T is called maturing date or expiring date

- At any time t, the value of B_t is

$$B_t = Fe^{-r(T-t)} = B_0 e^{rt},$$



Coupon bond

- With a coupon bond one receives not only the amount F at time T but also a sequence of payments, called coupons, during the life of the bond.
- The price of the bond is the total present value,

$$B_0 = \sum_{n=1}^{N} e^{-rt_n} C_n + F e^{-rT}$$



Internal Rate of Return

- Internal rate of return is a measure used to evaluate a sequence of financial events.
- used to rank projects that involve future cash flows:



- Consider the case that returns, for an initial investment of $P > 0$, an amount $A_n \geq 0$ at the end of periods $n = 1, 2, \ldots, N$.
- The internal rate of return(IRR) of the investment P is defined as the periodic interest rate i for which the present value of the sequence returns equals the initial payment P.
- Thus i

$$P = \sum_{n=1}^{N} A_n (1+i)^{-n} \tag{11}$$



- (11) has unique solution $i > -1$,
- Let

$$R(i) = \sum_{n=1}^{N} A_n (1+i)^{-n}$$

- R is continuous on $(-1, \infty)$

$$\lim_{i \to \infty} R(i) = 0 \quad \text{and} \quad \lim_{i \to -1^+} R(i) = \infty.$$

- Since $P > 0$, the intermediate value Theorem guarantees as solution $i > -1$ of the equation $R(i) = P$,
- Since R is strictly decreasing, the solution is unique



Remark 1

Because R is decreasing, $i > 0$ iff $R(i) < R(0)$ i.e., iff $P < \sum_{n=1}^{N} A_n$. Thus for positive rate of return, the sum of the payoffs is less than the initial investment.

